KERALA POLYTECHNIC · REVISION 2021

Introduction to Python

Complete student lesson for course code **1257**.

Revision 2021 Semester 1 Program Basic 4 modules

Course snapshot

Scheme: 3 (L:1, T:0, P:2)

Credits: 2

Contact: 45 periods per semester

Module hours listed: 42

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

Course Dashboard

Course code

1257

Semester

1

Credits

2

Teaching scheme

3 (L:1, T:0, P:2)

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Python Syntax, Operator Precedence and Control Flow	10	CO1
2	Functions and Strings	12	CO2
3	Lists, Dictionaries, Tuples and Sets	10	CO3
4	Files and Exception Handling	10	CO4

Prerequisites

- Basic understanding of computer terminologies (High School).

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO–PO Information

Official course objectives

1. To provide a hands-on experience in programming concepts using Python Programming Language. Students will be able to develop programs using arrays, functions, and files.
2. To enhance the problem-solving skills of students.

Course Outcomes

1. Apply the fundamental Python syntax and semantics and be fluent in the use of Python control flow statements.
2. Develop proficiency in the handling of strings and functions.
3. Construct Python programs by utilizing the data structures like lists, dictionaries, tuples and sets.
4. Identify the commonly used operations involving file systems and exception handling.

CO–PO mapping as supplied

CO1: PO1 strong (3), PO4 strong (3).
CO2: PO1 strong (3), PO4 strong (3).
CO3: PO1 strong (3), PO4 strong (3).
CO4: PO1 strong (3), PO4 strong (3).
Legend: 3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped.

Complete Syllabus Coverage

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	MO1.1	Operator precedence	2
module-1	MO1.2	Decision control structures	4
module-1	MO1.3	Loop control structures	4
module-2	MO2.1	Functions: swapping values and checking odd/even	3

Module	Topic code	Topic	Hours
module-2	MO2.3	Default arguments	3
module-2	MO2.3	Strings	6
module-3	MO3.1	Lists	3
module-3	MO3.2	Dictionaries	3
module-3	MO3.3	Lists: nested and practical operations	3
module-3	MO3.4	Sets	1
module-4	MO4.1	Exception handling	4
module-4	MO4.2	Create, open and close a file	3
module-4	MO4.3	Append a file and related functions	3

Learning Roadmap

1. Python Syntax, Operator Precedence and Control Flow
CO1 · 10 hours

2. Functions and Strings
CO2 · 12 hours

3. Lists, Dictionaries, Tuples and Sets
CO3 · 10 hours

4. Files and Exception Handling
CO4 · 10 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Python Syntax, Operator Precedence and Control Flow

This module establishes the grammar of a Python program and then uses decisions and repetition to express algorithms. The emphasis is on writing small, testable programs rather than memorising isolated syntax.

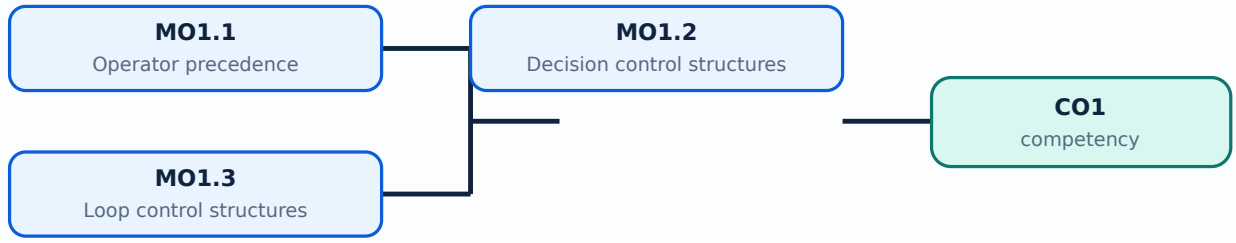
Hours: 10
CO1 · Bloom level: Applying

Learning goals

- Recognise literals, variables, expressions, indentation, input/output, and operator precedence.
- Select an appropriate decision structure for a problem.
- Use for and while loops with correct initialisation, condition, update, and termination.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

MO1.1 — Operator precedence (2 hours)

Concept and explanation

Python evaluates an expression according to a precedence hierarchy. Parentheses are evaluated first, followed by exponentiation, unary operators, multiplication/division/floor division/modulus, addition/subtraction, comparisons, membership or identity tests, and Boolean not, and, or. When an expression is not immediately clear, parentheses are the safest way to communicate the intended order. Assignment is a statement rather than an arithmetic operator.

Malayalam support: Python-□ expression evaluate □□□□□□□□□□□ operator precedence □□□ order □□□□□□□□□□□□□□□□□□. □□□□□ □□□□□□□□□□ brackets □□□□□□□□□□□□□□□ □□□□□□□ □□□□□□□□□□□□.

Study points

- Use meaningful variable names; distinguish / from //; test expressions with small values; avoid depending on remembered precedence for long expressions.

Engineering / workplace application: Operator precedence is important in marks calculation, sensor thresholds, billing expressions, and validation logic.

Illustrative example: Given a=8 and b=3, `a + b * 2` gives 14 because multiplication is evaluated first; `(a+b)*2` gives 22.

Common mistake: Writing a*b+c as a*(b+c) without checking the intended formula changes the result.

MO1.2 — Decision control structures (4 hours)

Concept and explanation

Decision structures choose a path based on a Boolean condition. Python uses if, elif, and else with indentation to delimit blocks. Conditions may combine comparisons with and, or, and not; nested decisions should be used only when they make the logic clearer. A complete program should consider equality, range boundaries, and invalid input.

Malayalam support: Condition □□□□□□□□ □□□□□□□□□ □□□□□□□□□ □□□□□□□□□□□□□□□ program □□□ □□□ □□□□□□ □□□□□□ if/elif/else □□□□□□□□□□□□□□□□□□. Indentation Python-□ block □□□□□□□□□□□□□□□□□ □□□□□□□□□ □□□□□□□□□□□.

Study points

- Convert the problem statement into conditions; put mutually exclusive cases in a clear order; include an else or validation branch where appropriate; test boundary values.

Engineering / workplace application: Grade classification, tax slabs, pass/fail decisions, menu selection, and machine interlocks use decision logic.

Illustrative example: For marks m, `if m>=75: distinction; elif m>=50: pass; else: fail` evaluates the highest applicable band first.

Common mistake: Using `if x > 10` when the requirement is `x >= 10` loses the boundary case.

MO1.3 — Loop control structures (4 hours)

Concept and explanation

A loop repeats a block while a pattern or condition remains valid. A for loop is suitable for iterating through a known sequence or range; a while loop is suitable when repetition depends on a changing condition. Initialise the control variable, update it, and make the exit condition reachable. break stops a loop, continue skips to the next iteration, and an else clause can run when a loop ends normally.

Malayalam support: Loop □□□□□□□□□□ □□□ □□□□□ □□ □□□ □□□□□□□□□□□□. for □□□□□□□□□□ sequence-□□□□□□□□□ □□□□□□□□□□□; while condition □□□□□□□□□□□□ □□□□□□□□□□□□□□. Exit condition □□□□□□□□□□ infinite loop □□□□□□□□.

Study points

- Trace one iteration in a table; keep loop bodies short; use range endpoints carefully; prevent infinite loops; validate input before repeating.

Engineering / workplace application: Counters, data sampling, menu-driven programs, and repeated file records are natural loop applications.

Illustrative example: A loop from 1 to 5 inclusive is `for i in range(1, 6):`; the stop value in range is excluded.

Common mistake: Forgetting to update a while-loop variable causes an infinite loop; using range(1,5) when 5 must be included also causes an error.

Module summary: Syntax supports expressions; decisions select one path; loops express controlled repetition.

OFFICIAL MODULE 2

Functions and Strings

Functions divide a solution into named, reusable units. Strings provide a safe way to represent and process text; together they support readable programs for validation, searching, formatting, and small automation tasks.

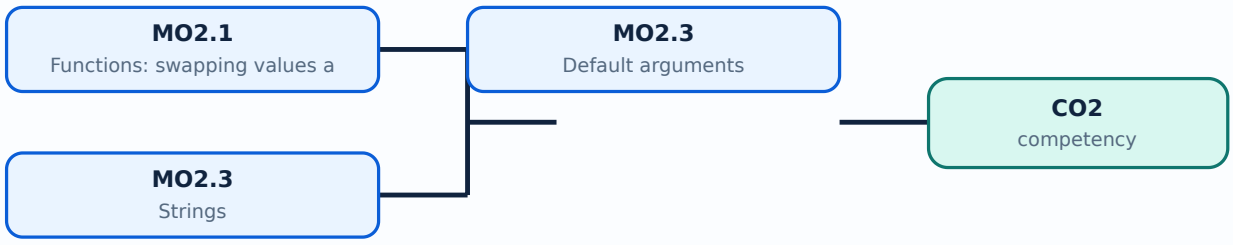
Hours: 12
CO2 · Bloom level: Applying

Learning goals

- Define and call functions with parameters and return values.
- Use default arguments and understand local scope.
- Apply common string operations without confusing indexing, slicing, and mutation.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

MO2.1 — Functions: swapping values and checking odd/even (3 hours)

Concept and explanation

A function is defined with `def`, accepts parameters, performs a focused task, and may return a result. In Python, assignment can swap two variables as `a, b = b, a`, without a temporary variable. An odd/even function can use the remainder operator: n % 2 is zero for an even integer. The function should be tested with positive, negative, and zero inputs.

Malayalam support: Function □□□□□□ □□□ □□□□ □□□□□□ □□□□□□□□□ reusable code block □□□. `a,b=b,a` □□□□□□□□□□ variables swap □□□□□□□. `% 2` □□□□□□□□□□ number odd □□□ even □□□ □□□□□□□□□□□□.

Study points

- Keep one responsibility per function; choose descriptive names; return values instead of printing when the caller may need the result; document assumptions.

Engineering / workplace application: Functions are used for reusable validation, payroll rules, conversion routines, and laboratory data processing.

Illustrative example: `def is_even(n): return n % 2 == 0` returns a Boolean value that can be used by another function.

Common mistake: Printing inside a function when the caller expects a returned value makes composition difficult.

MO2.3 — Default arguments (3 hours)

Concept and explanation

A default argument supplies a value when the caller omits that parameter. It is written after required parameters, for example `def greet(name, message="Welcome")`. Default values should be immutable or deliberately handled; a mutable default list can retain data between calls and cause surprising behaviour.

Malayalam support: Argument □□□□□□□□□□□□ □□□□□□□□□□□□□ □□□□□□□□□ □□□□□ value □□□ default argument. Required argument-□□□□□□□ □□□□ default parameter □□□□□□.

Study points

- Call functions both with and without the optional value; place required parameters before defaults; avoid mutable defaults such as [] unless intentionally managed.

Engineering / workplace application: Configuration functions, report formatting, and optional search limits benefit from defaults.

Illustrative example: `def area(length, breadth=1): return length*breadth` calculates a rectangle area or a strip area when breadth is omitted.

Common mistake: Defining a required parameter after a default parameter raises a syntax error.

MO2.3 — Strings (6 hours)

Concept and explanation

A string is an ordered, immutable sequence of characters. Indexing starts at zero, negative indices count from the end, and slicing uses start:stop:step with an exclusive stop. Common methods include lower, upper, strip, split, join, replace, find, and count. Concatenation and f-strings format text; input arrives as a string and may need conversion.

Malayalam support: String □□□□□□ characters-□□□□ immutable sequence □□□. Index 0 □□□□ □□□□□□□□; slicing-□ stop position □□□□□□□□□□□. Input-□□ numeric operation □□□□□□□ int/float □□□ convert □□□□□□□.

Study points

- Check empty strings; distinguish `find` returning -1 from an exception; use strip before comparing user input; prefer f-strings for readable output.

Engineering / workplace application: Names, commands, CSV fields, passwords, addresses, and report labels are all processed as strings.

Illustrative example: For `s="Python"`, `s[1:4]` is `yth`; `"-".join(["A","B"])` produces `A-B`.

Common mistake: Trying to change `s[0]` directly fails because strings are immutable; create a new string instead.

Module summary: Functions make solutions reusable; strings make text processing predictable and testable.

OFFICIAL MODULE 3

Lists, Dictionaries, Tuples and Sets

Python collections store groups of values. The correct data structure depends on whether order, mutability, duplicate values, or key-based access is most important.

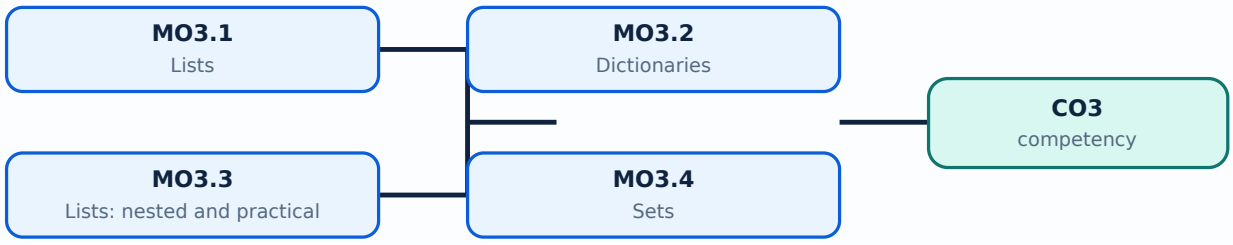
Hours: 10
CO3 · Bloom level: Applying

Learning goals

- Create, index, slice, update, and traverse lists.
- Use dictionaries for key-value records and tuples for fixed grouped values.
- Use sets for unique items and set operations.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

MO3.1 — Lists (3 hours)

Concept and explanation

A list is an ordered, mutable collection written with square brackets. It may contain mixed data types, supports indexing and slicing, and provides append, extend, insert, remove, pop, sort, and reverse operations. Iterating over a list is clearer than repeatedly writing separate variables. Copying with `b=a` aliases the same list; `a.copy()` creates a shallow copy.

Malayalam support: List □□□□□□ order □□□□□□□□□□□□□, □□□□□□□ □□□□□□□□ collection □□□. append □□□ item □□□□□□□□; extend □□ items □□□□□□□□.

Study points

- Select list when order and change are required; check index range; use meaningful iteration variables; do not modify a list unpredictably while iterating.

Engineering / workplace application: Marks, inventory items, sensor readings, and menu items are naturally represented as lists.

Illustrative example: A list of marks can be averaged with `sum(marks)/len(marks)` after checking that the list is not empty.

Common mistake: `b=a` independent copy □□□□; b-□□□ □□□□□□ □□□□□□□□□□ a-□□□ □□□□□.

MO3.2 — Dictionaries (3 hours)

Concept and explanation

A dictionary stores key-value pairs. Keys must be hashable and unique; values may be any suitable object. Access can use `d[key]` when the key must exist or `d.get(key, default)` when absence is expected. Items, keys, and values views support iteration. Dictionaries are useful for records and fast lookup.

Malayalam support: Dictionary-□ key □□□□□□□□□□ value □□□□□□□□□□. Key unique □□□□□□□□□□. Key □□□□□□□□□□□□□ `get()` □□□□□□□□□□□□□□□□ safer □□□.

Study points

- Use stable, descriptive keys; validate required keys; avoid duplicate keys in literals; iterate with items when both key and value are needed.

Engineering / workplace application: Student records, product prices, configuration settings, and frequency counts use dictionaries.

Illustrative example: `price={"pen":10,"book":50}` gives 50 for `price["book"]`; `price.get("bag",0)` safely returns zero if bag is absent.

Common mistake: Accessing a missing key with brackets raises KeyError; `get` with a default can handle optional data.

MO3.3 — Lists: nested and practical operations (3 hours)

Concept and explanation

The supplied outline repeats the word lists for MO3.3. This lesson preserves that official label and uses the topic for nested lists, list comprehensions, filtering, and practical traversal. A nested list is a list whose elements are themselves lists; access requires one index per level.

Malayalam support: □ syllabus-□ MO3.3-□ lists □□□□□□□ □□□□□□□□□□□□□□. □□□□□ nested lists, filtering, practical traversal □□□□ □□□□□□□ □□□□□□□□.

Study points

- Represent tables as nested lists; verify row lengths; use comprehensions only when they remain readable; keep transformations separate from input.

Engineering / workplace application: A marks table, pixel grid, timetable, or matrix can be modelled as a nested list.

Illustrative example: For `rows=[[1,2],[3,4]]`, `rows[1][0]` is 3; `[x*x for x in range(4)]` creates square values.

Common mistake: Assuming every row has the same length without checking causes IndexError on irregular data.

MO3.4 — Sets (1 hours)

Concept and explanation

A set is an unordered collection of unique hashable elements. It supports add, remove/discard, union, intersection, difference, and membership testing. Sets are ideal for eliminating duplicates, but they do not preserve a meaningful display order.

> **Malayalam support:** Set-□ duplicate values □□□□□□□□□□□□□□; order-□□ □□□□□□□□□□□□□□. Union, intersection □□□□□□□□ operations □□□□□□□□□□ groups compare □□□□□□□□.

Study points

- Use a set for uniqueness or membership; use discard when missing items are acceptable; convert to a sorted list when ordered output is needed.

> **Engineering / workplace application:** Removing duplicate registration numbers or finding common skills between teams are set applications.

Illustrative example: `set([1,1,2,3])` becomes `{1,2,3}`; `{1,2}&{2,3}` gives `{2}`.

> **Common mistake:** Indexing a set such as s[0] is invalid because sets are not sequences.

> **Module summary:** Choose collections by required order, mutability, uniqueness, and access method.

OFFICIAL MODULE 4

Files and Exception Handling

Programs become useful when they can preserve information and recover from expected errors. Files provide persistent storage; exceptions let a program handle failures without hiding genuine defects.

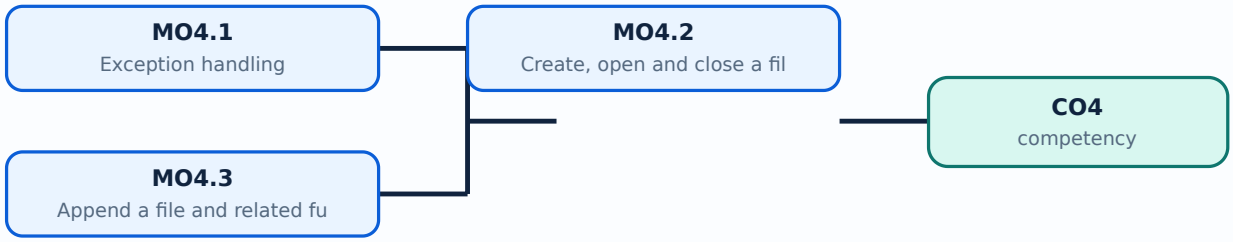
Hours: 10
CO4 · Bloom level: Applying

Learning goals

- Use try, except, else, and finally appropriately.
- Create, open, read, write, append, and close text files.
- Use context managers and validate file operations.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

MO4.1 — Exception handling (4 hours)

Concept and explanation

An exception is an event that interrupts normal execution, such as ValueError, TypeError, IndexError, KeyError, or FileNotFoundError. A try block contains risky operations; except handles a known failure; else runs when no exception occurs; finally runs for cleanup. Catch specific exceptions rather than using a bare except, and do not use exception handling to mask incorrect logic.

Malayalam support: Exception □□□□□□□□□□□□□□□ program □□□□□□□□□□□ □□□□□□□ expected error handle □□□□□□□ try/except □□□□□□□□□□□□□□□. Specific exception catch □□□□□□□□□□□□□□ □□□□□□□.

Study points

- Validate before conversion; catch only expected exceptions; show a useful message; preserve cleanup in finally or a context manager; never expose sensitive data in error messages.

Engineering / workplace application: Input validation, file access, network requests, and database operations all need controlled error handling.

Illustrative example: For `int(input())`, catch ValueError and ask for a whole number again; do not catch every possible exception as ValueError.

Common mistake: A bare `except` can hide programming errors and make debugging difficult.

MO4.2 — Create, open and close a file (3 hours)

Concept and explanation

A file is persistent storage identified by a path. `open(path, mode, encoding="utf-8")` returns a file object; common modes are r for read, w for overwrite/create, x for exclusive creation, and a for append. The `with open(...) as f:` context manager closes the file automatically, even when an exception occurs.

Malayalam support: File open □□□□□□□□□□□□ mode □□□□ □□□□□□□□□□□: w □□□ content overwrite □□□□□□□□, a □□□□□□ □□□□□□□□□, r □□□□□□□□□□. `with` □□□□□□□□□□□□ file automatic □□□ close □□□□□□□□.

Study points

- Use UTF-8 for text; use a relative or validated path; never use w when data must be preserved; close resources deterministically.

Engineering / workplace application: Logs, student records, configuration files, and generated reports require file operations.

Illustrative example: `with open("notes.txt","w",encoding="utf-8") as f: f.write("Hello ")` creates or replaces the text file and closes it.

Common mistake: Opening with w accidentally erases the file; assuming a relative path refers to the same folder in every environment is unsafe.

MO4.3 — Append a file and related functions (3 hours)

Concept and explanation

Appending preserves existing content and writes new data at the end. `read`, `readline`, `readlines`, iteration, `write`, and `writelines` are common operations. A robust program checks file existence, uses newline conventions consistently, and verifies the result. For structured data, a simple delimiter or a standard format should be chosen deliberately.

Malayalam support: Append mode □□□ content □□□□□□□□□□ □□□□□ record □□□□□□ □□□□□□□□. □□□□□□□□□□□ □□□□□□□□ newline, encoding, format □□□□□□ □□□□□□□□□□□□□.

Study points

- Append one complete record per line; ensure separators are present; read back after writing during testing; use a clear file format.

Engineering / workplace application: Daily transaction logs and attendance records are typical append-only files.

Illustrative example: `with open("log.txt","a",encoding="utf-8") as f: f.write("2026-08-15,OK ")` appends one record.

Common mistake: Calling read after the file pointer has reached the end returns an empty string unless seek(0) is used.

Module summary: A reliable program validates input, manages resources, and preserves data through deliberate file modes.

Formula and Notation Bank

Remainder test

```
n % 2 == 0
```

Meaning: n is the integer; % returns the remainder. The result is Boolean and has no physical unit.

Average of a non-empty list

```
average = sum(values) / len(values)
```

Meaning: the numerator is the total and the denominator is the item count.

Diagram Library for Rapid Revision

e module to view its labelled learning-map diagram.

Module 2: Functions and

elled learning-map diagram.

Module 3: Lists,

iew its labelled learning-map diagram.

Module 4: Files and

Open the module to view its labelled learning-map diagram.

Official Activities and Practical Requirements

Official activities / self-learning

- Write and run one small program after each topic.
- Maintain a practical record containing problem statement, algorithm, source code, test data, output, and short reflection.
- Test boundary cases such as zero, negative values, empty strings, and empty collections.

Practical requirements / equipment

- A Python 3 interpreter or IDLE/terminal environment; the syllabus does not prescribe a specific software package.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- The PDF lists a Lab Exam of 3 hours but does not provide a separate CIE/ESE mark distribution or question pattern. This lesson therefore treats the practice paper as non-official.
- Use the practical record, executable programs, output verification, and viva preparation as study evidence; the exact institutional marking scheme must be confirmed with the department.

Module-wise Questions and Answers

module-1 — Python Syntax, Operator Precedence and Control Flow

1. Explain Operator precedence. [3 marks]

Python evaluates an expression according to a precedence hierarchy. Parentheses are evaluated first, followed by exponentiation, unary operators, multiplication/division/floor division/modulus, addition/subtraction, comparisons, membership or identity tests, and Boolean not, and, or. When an expression is not immediately clear, parentheses are the safest way to communicate the intended order. Assignment is a statement rather than an arithmetic operator.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Decision control structures. [3 marks]

Decision structures choose a path based on a Boolean condition. Python uses if, elif, and else with indentation to delimit blocks. Conditions may combine comparisons with and, or, and not; nested decisions should be used only when they make the logic clearer. A complete program should consider equality, range boundaries, and invalid input.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Loop control structures. [3 marks]

A loop repeats a block while a pattern or condition remains valid. A for loop is suitable for iterating through a known sequence or range; a while loop is suitable when repetition depends on a changing condition. Initialise the control variable, update it, and make the exit condition reachable. break stops a loop, continue skips to the next iteration, and an else clause can run when a loop ends normally.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — Functions and Strings

4. Explain Functions: swapping values and checking odd/even. [3 marks]

A function is defined with `def`, accepts parameters, performs a focused task, and may return a result. In Python, assignment can swap two variables as `a, b = b, a`, without a temporary variable. An odd/even function can use the remainder operator: n % 2 is zero for an even integer. The function should be tested with positive, negative, and zero inputs.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

5. Explain Default arguments. [3 marks]

A default argument supplies a value when the caller omits that parameter. It is written after required parameters, for example `def greet(name, message="Welcome")`. Default values should be immutable or deliberately handled; a mutable default list can retain data between calls and cause surprising behaviour.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Strings. [3 marks]

A string is an ordered, immutable sequence of characters. Indexing starts at zero, negative indices count from the end, and slicing uses start:stop:step with an exclusive stop. Common methods include lower, upper, strip, split, join, replace, find, and count. Concatenation and f-strings format text; input arrives as a string and may need conversion.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Lists, Dictionaries, Tuples and Sets

7. Explain Lists. [3 marks]

A list is an ordered, mutable collection written with square brackets. It may contain mixed data types, supports indexing and slicing, and provides append, extend, insert, remove, pop, sort, and reverse operations. Iterating over a list is clearer than repeatedly writing separate variables. Copying with `b=a` aliases the same list; `a.copy()` creates a shallow copy.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Dictionaries. [3 marks]

A dictionary stores key-value pairs. Keys must be hashable and unique; values may be any suitable object. Access can use `d[key]` when the key must exist or `d.get(key, default)` when absence is expected. Items, keys, and values views support iteration. Dictionaries are useful for records and fast lookup.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

9. Explain Lists: nested and practical operations. [3 marks]

The supplied outline repeats the word lists for MO3.3. This lesson preserves that official label and uses the topic for nested lists, list comprehensions, filtering, and practical traversal. A nested list is a list whose elements are themselves lists; access requires one index per level.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Sets. [3 marks]

A set is an unordered collection of unique hashable elements. It supports add, remove/discard, union, intersection, difference, and membership testing. Sets are ideal for

eliminating duplicates, but they do not preserve a meaningful display order.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — Files and Exception Handling

11. Explain Exception handling. [3 marks]

An exception is an event that interrupts normal execution, such as ValueError, TypeError, IndexError, KeyError, or FileNotFoundError. A try block contains risky operations; except handles a known failure; else runs when no exception occurs; finally runs for cleanup. Catch specific exceptions rather than using a bare except, and do not use exception handling to mask incorrect logic.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Create, open and close a file. [3 marks]

A file is persistent storage identified by a path. `open(path, mode, encoding="utf-8")` returns a file object; common modes are r for read, w for overwrite/create, x for exclusive creation, and a for append. The `with open(...) as f:` context manager closes the file automatically, even when an exception occurs.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

13. Explain Append a file and related functions. [3 marks]

Appending preserves existing content and writes new data at the end. `read`, `readline`, `readlines`, iteration, `write`, and `writelines` are common operations. A robust program checks file existence, uses newline conventions consistently, and verifies the result. For structured data, a simple delimiter or a standard format should be chosen deliberately.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Operator precedence* with one relevant application. (5 marks)
2. **2.** Define or explain *Decision control structures* with one relevant application. (5 marks)
3. **3.** Define or explain *Functions: swapping values and checking odd/even* with one relevant application. (5 marks)
4. **4.** Define or explain *Default arguments* with one relevant application. (5 marks)
5. **5.** Define or explain *Lists* with one relevant application. (5 marks)
6. **6.** Define or explain *Dictionaries* with one relevant application. (5 marks)
7. **7.** Define or explain *Exception handling* with one relevant application. (5 marks)
8. **8.** Define or explain *Create, open and close a file* with one relevant application. (5 marks)

> **Writing advice:** Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Python Syntax, Operator Precedence and Control Flow:** Syntax supports expressions; decisions select one path; loops express controlled repetition.
- **Functions and Strings:** Functions make solutions reusable; strings make text processing predictable and testable.
- **Lists, Dictionaries, Tuples and Sets:** Choose collections by required order, mutability, uniqueness, and access method.
- **Files and Exception Handling:** A reliable program validates input, manages resources, and preserves data through deliberate file modes.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Operator precedence	Python evaluates an expression according to a precedence hierarchy. Parentheses are evaluated first, followed by exponentiation, unary operators, multiplication/division/floor division/modulus, addition/subtraction, comparisons, membership or identity tests, a
Decision control structures	Decision structures choose a path based on a Boolean condition. Python uses if, elif, and else with indentation to delimit blocks. Conditions may combine comparisons with and, or, and not; nested decisions should be used only when they make the logic clearer.
Loop control structures	A loop repeats a block while a pattern or condition remains valid. A for loop is suitable for iterating through a known sequence or range; a while loop is suitable when repetition depends on a changing condition. Initialise the control variable, update it, and
Functions: swapping values and checking odd/even	A function is defined with `def`, accepts parameters, performs a focused task, and may return a result. In Python, assignment can swap two variables as `a, b = b, a`, without a temporary variable. An odd/even function can use the remainder operator: n % 2 is z
Default arguments	A default argument supplies a value when the caller omits that parameter. It is written after required parameters, for example `def greet(name, message="Welcome")`. Default values should be immutable or deliberately handled; a mutable default list can retain d
Strings	A string is an ordered, immutable sequence of characters. Indexing starts at zero, negative indices count from the end, and slicing uses start:stop:step with an exclusive stop. Common methods include lower, upper, strip, split, join, replace, find, and count.
Lists	A list is an ordered, mutable collection written with square brackets. It may contain mixed data types, supports indexing and slicing, and provides append, extend, insert, remove, pop, sort, and reverse operations. Iterating over a list is clearer than repeate
Dictionaries	A dictionary stores key-value pairs. Keys must be hashable and unique; values may be any suitable object. Access can use `d[key]` when the key must exist or `d.get(key, default)` when absence is expected. Items, keys, and values views support iteration. Dictio

Term	Short meaning
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Sets	A set is an unordered collection of unique hashable elements. It supports add, remove/discard, union, intersection, difference, and membership testing. Sets are ideal for eliminating duplicates, but they do not preserve a meaningful display order.
Exception handling	An exception is an event that interrupts normal execution, such as ValueError, TypeError, IndexError, KeyError, or FileNotFoundError. A try block contains risky operations; except handles a known failure; else runs when no exception occurs; finally runs for cl
Create, open and close a file	A file is persistent storage identified by a path. `open(path, mode, encoding="utf-8")` returns a file object; common modes are r for read, w for overwrite/create, x for exclusive creation, and a for append. The `with open(...) as f:` context manager closes th
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Official References and Resources

References listed in the supplied syllabus

1. Gowrishankar S and Veena A, Introduction to Python Programming, 1st Edition, CRC Press/Taylor & Francis, 2018.
2. Python3 for Absolute Beginners, Tim Hall and J. P. Stacey.
3. Head First Python: A Brain-Friendly Guide, Paul Barry.

Official learning resources listed

- https://www.w3schools.com/python/
- https://www.tutorialspoint.com/python/
- https://www.geeksforgeeks.org/python-programming-language/

In-page Search

material.